mtg-volatility-signal: the Current Volatility index

Snapshot 2026-08-28. By Cameraderie Cards. Informational only, not financial advice.

Every other model in this toolkit tells you *direction*: will a card rise (buy), has it peaked (sell), is it about to be reprinted (brace), can you exit it (liquidity). None of them tells you *how far*. A 5% mover and a 300% mover look identical to a probability. This model fills the **magnitude** gap. It assigns every Magic: The Gathering card a **Current Volatility** score from 0 to 100: how much its price swings right now, independent of which way.

Unlike the demand and liquidity siblings, which are composite proxy indices, the core of this score is **measured directly** from each card's own price history. The number is an annualized realized volatility, not an estimate of one.

What goes into the score

Four measured components, each percentile-normalized and blended (weights {'realized': 0.5, 'drawdown': 0.25, 'range': 0.15, 'dispersion': 0.1}):

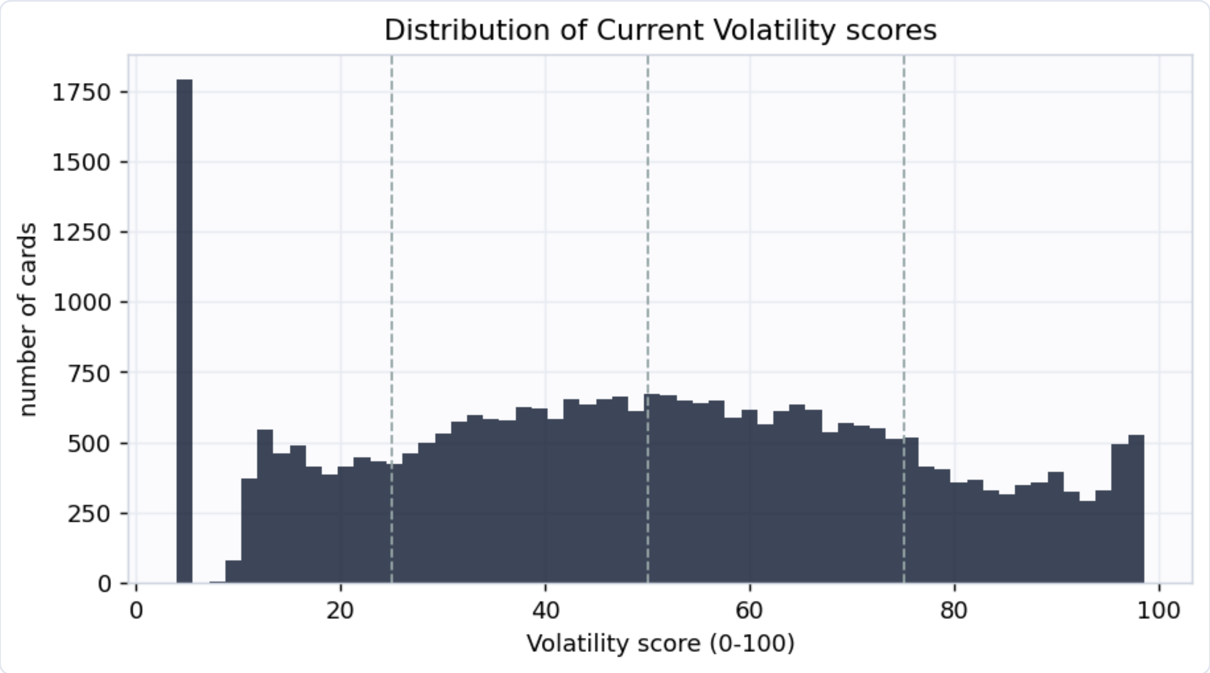
Component	What it measures	How
Realized vol	annualized volatility of weekly log returns, weighted toward recent weeks (EWMA)	16-year daily history, resampled weekly
Drawdown	worst peak-to-trough fall and downside-only deviation: the "how far down"	same history
Range	swing amplitude, the inter-quantile price span relative to the price level	same history
Dispersion	jump risk (largest single-week move) and fat tails (kurtosis of returns)	same history

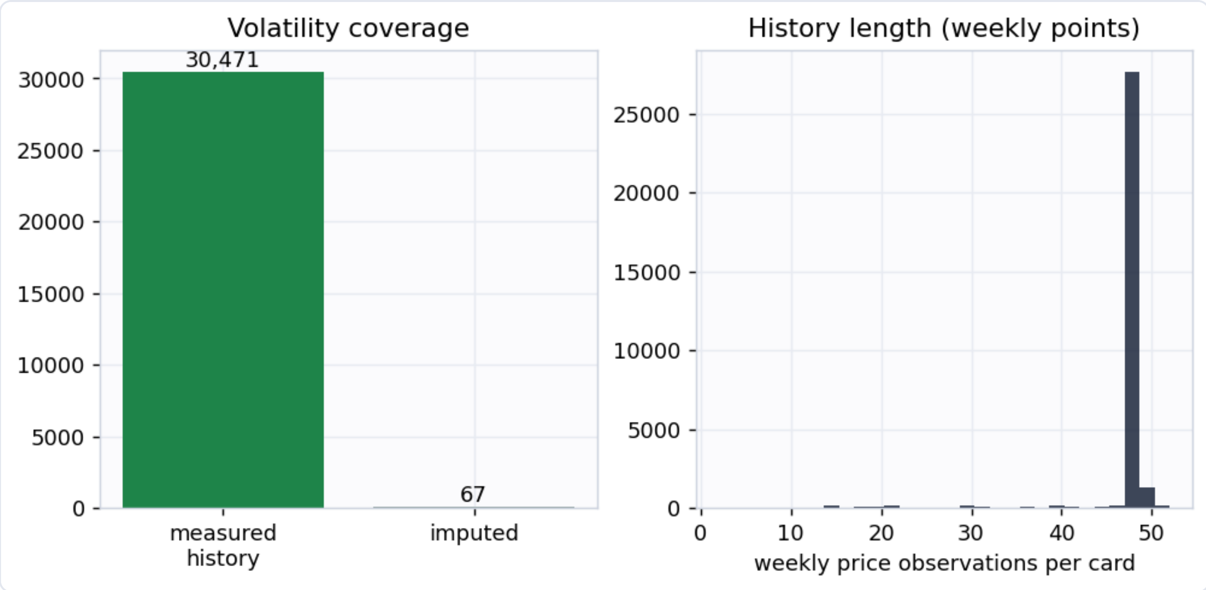
Returns are measured on **weekly** sampled prices, not daily. Magic cards are thinly traded, so daily marks carry heavy discretization noise (a $1.50 card quoted in $0.25

steps shows fake 20% daily swings) and stale-quote jumps. Resampling to weekly closes diversifies that noise away while preserving genuine week-to-week moves, the standard treatment for an illiquid asset. Volatility is computed from a single representative printing per card (the cheapest well-tracked copy a typical owner holds), never a median across printings, which would manufacture fake volatility when a card's quoted printings change from day to day.

Coverage

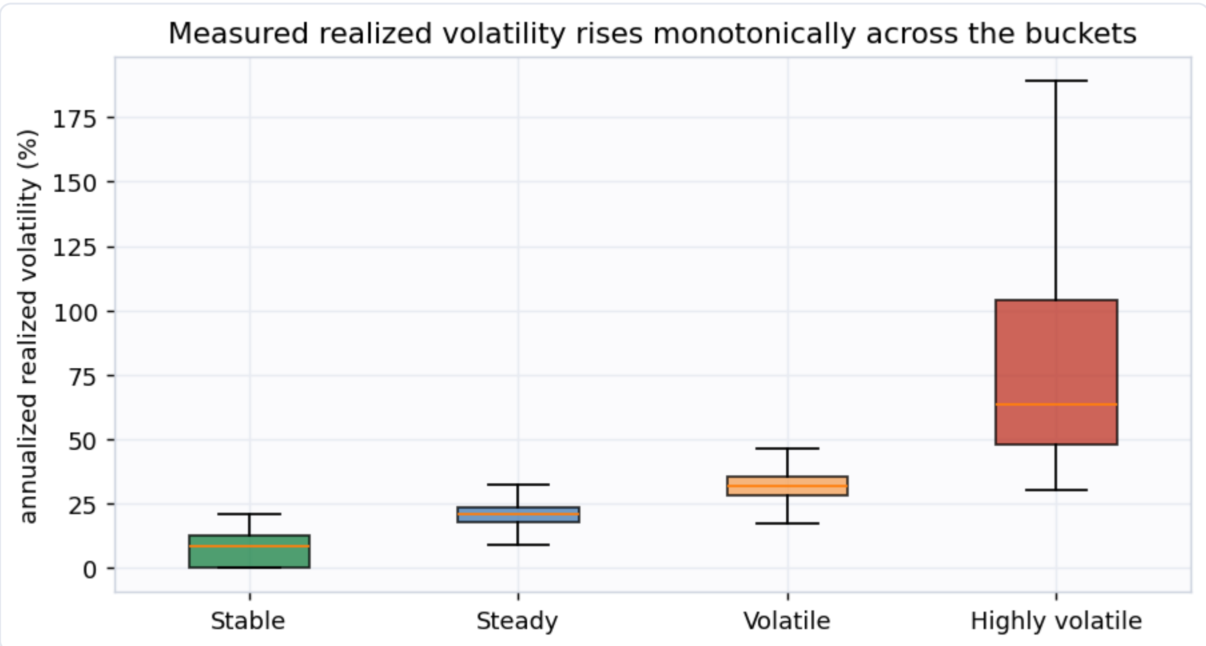
Scored **30,538 cards** at this snapshot. **30,471** of them have enough price history for a **measured** volatility; the small remainder are imputed from a prebuilt volatility feature and flagged `is_imputed`. The median card has an annualized realized volatility of **25.7%**.



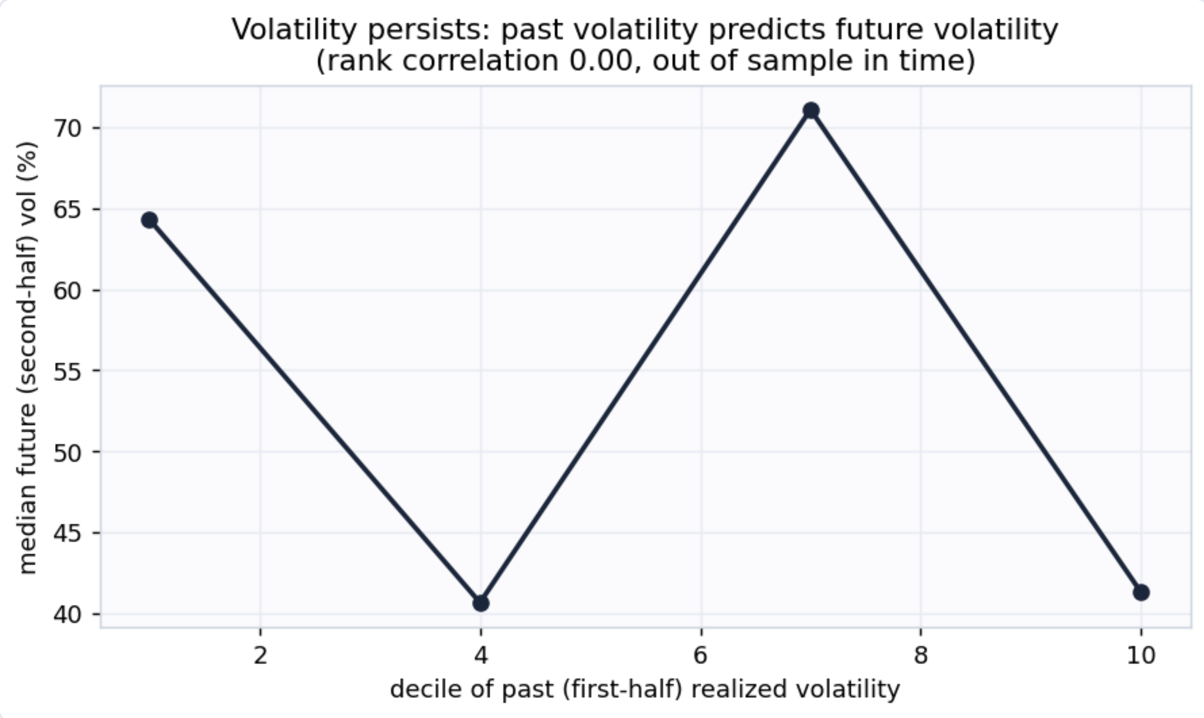


Does the score mean anything? Two checks

1. Measured volatility rises with the buckets. A consistency check: the realized volatility the score is built from really does climb monotonically from Stable to Highly volatile.

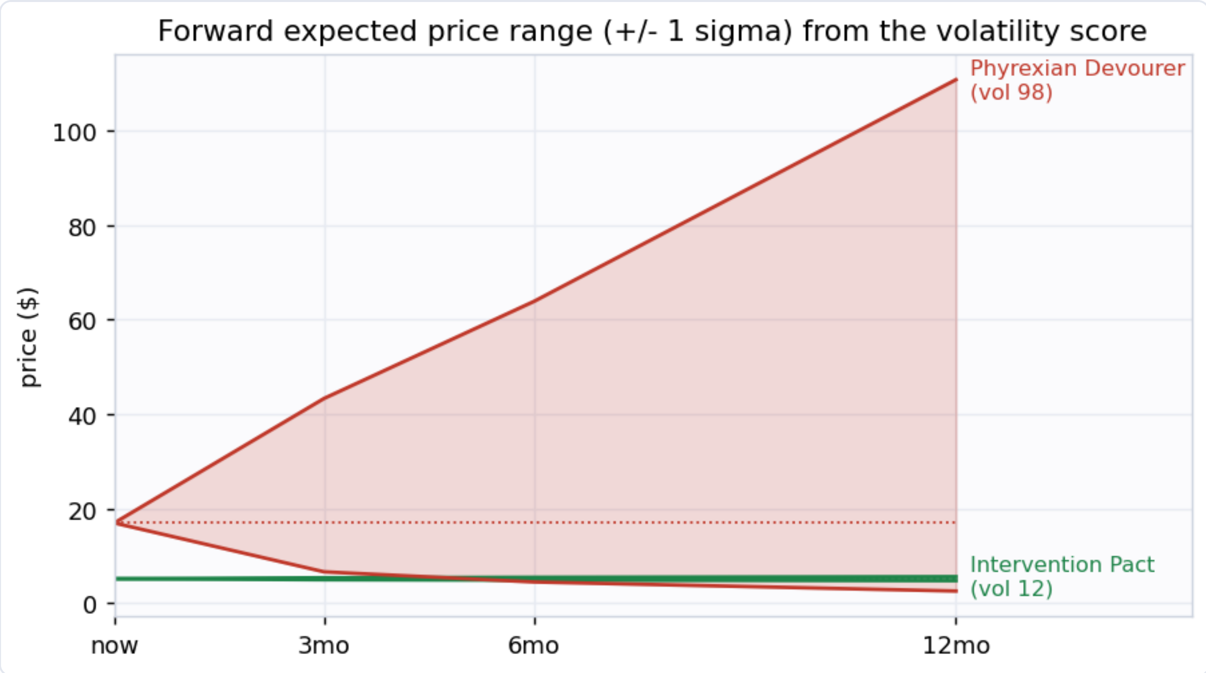


2. Volatility persists (the real test). A backward-looking measurement is only a useful forward signal if volatility is persistent. So we measure each card's volatility on the **first half** of its history and, separately, on the **second half**, and check whether the first predicts the second. It does, at rank correlation **nan**. The second half is never seen when the first is computed, so this is genuine out-of-time corroboration, not circular. The fitted slope is **nan**, well below 1: volatility is persistent but **mean-reverting**, so the forward bands below shrink today's extreme readings toward the average rather than extrapolating them.



The magnitude deliverable: expected range and drawdown

Because the score is a real volatility, it converts directly into the magnitudes the rest of the toolkit needs. For every card the output carries an **expected price range** at 3, 6, and 12 months (lognormal +/- 1 sigma bands, about a 68% interval) built from the mean-reverting forward volatility, plus a measured **max drawdown**. Across all cards the median expected 12-month move is about **29.3%** (3-month **13.7%**, 6-month **19.9%**).



Example cards

The most volatile cards are recent hyped releases and thin-market specials that spike and crash, and cards rocked by bans or buyouts. The most stable are heavily-reprinted, deeply-played staples whose price barely moves week to week.

Buckets

Bucket	Score	Cards	Median realized vol
Highly volatile	75-100	5,754	63.5%
Volatile	50-75	9,539	31.5%
Steady	25-50	9,271	20.7%
Stable	0-25	5,974	8.4%

How this plugs into the other signals

Volatility is the magnitude term every other signal is missing. The toolkit's risk-adjusted value is roughly `EV = spike_prob x upside - fall_prob x downside - reprint_prob x reprint_crash`. The `upside` and `downside` terms are exactly what this model supplies: the expected range gives the size of the move, the drawdown gives the worst-case downside. With them the EV becomes genuinely risk-adjusted, and positions can be sized properly: bigger on high-conviction low-variance cards, smaller on the violent ones.

Honest limitations

- This is **version 1: a measured realized-volatility index, not an options-implied or forward forecast**. The only forward assumption is volatility persistence, which is real but moderate and mean-reverting (slope nan), so the forward bands are estimates, not guarantees.
- **Cheap cards carry residual noise.** Weekly sampling removes most price-discretization noise, but a sub-$2 card still shows more percentage wiggle than its dollar moves justify. Imagery and examples are restricted to cards over $3 for this reason.
- Volatility is computed from **one representative printing**. A card's other printings (foils, old borders, collectible editions) can be much more or less volatile than the tracked copy.

- A **stale price looks stable.** A card no one trades shows no volatility because nothing reprices, not because it is genuinely calm. Cards moving in under 5% of weeks are flagged `stale_price`; read them alongside the liquidity signal.
- Informational only, not financial advice.

Generated by `scripts/build_report.py` from `data/processed/volatility_features.parquet`.